Jacob Ford

Senior Data Scientist and Tech Lead

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Summary

Senior data scientist and technical lead with 8+ years across energy, machine learning, and applied AI. Currently manages a team of 10 building production AI/ML systems and agentic developer tooling for battery energy-storage operations at FlexGen. Coordinate across disparate teams to identify AI and ML solutions to requisite problems. Businss impact, especially in LLM applications, is a focus, with a record of designing production systems and proving measurable business impact, including a 40% gain in engineering efficiency from agentic workflows in 6 months.

Technical Skills

AI & ML	LLMs & RAG, MCPs, Agentic systems, MLOps, Model design and development
Cloud & Data	AWS (Bedrock and Agentcore, S3 Vectors, SageMaker), Snowflake and Databricks, DataRobot
Leadership	Technical Project Leadership, ML Product Management, Agile, AI Evaluation

Experience

Senior Data Scientist and Tech Lead, *FlexGen* May 2025 – Present — New York, NY

- Lead a team of 10 data scientists and engineers building ML and AI systems for battery-storage operations.
- Architected and shipped ROC Assistant, an agentic fault analysis, clearance and preventative system built on AWS AgentCore Bedrock leading to 40% decrease in operator time and adoption of proactive maintenance.
- Led a cross-team task force that built an internal MCP server letting teams across the organization register and share agentic tools — establishing standards, contribution guidelines, and PR-review processes for org-wide adoption.

Senior Data Scientist, *Solstice* March 2022 – May 2025 — Cambridge, MA

- Secured $1M from Google to develop EnergyScore, an ML-based green-financing qualification tool, leading technical project management across data collection, model design, and MLOps strategy.
- Led development of a RAG chatbot for solar policy using Snowflake Cortex and custom embeddings, improving internal search across regulatory texts.
- Designed a custom churn-prediction model that segmented customers into risk clusters, driving a 39% decrease in churn among high-risk customers.
- Built and deployed a PyTorch document-classification model via AWS SageMaker, ECR, and Lambda (Docker-based serving), cutting manual review of LMI documents by 70%.

Transportation Modeler, *Triangle West TPO* February 2020 – March 2022 — Durham, NC

- Maintained a transportation model predicting multimodal travel volume across highway, transit, and pedestrian networks.
- Applied iterative proportional fitting, neural networks, and decision trees to estimate trip counts and purposes.

Program Evaluator, *North Carolina General Assembly* July 2018 – February 2020 — Raleigh, NC

- Conducted state policy evaluations and recommended programmatic solutions within the legislature's in-house watchdog agency.
- Identified $3.7M in unnecessary state spending stemming from an errant matching mechanism for hurricane-relief funding.

Education

Master of Public Policy, Georgetown University May 2018
Editor-in-chief, Georgetown Public Policy Review; Baker Innovation Scholar; Research Fellow, CROCUS

Bachelor of Science in Mathematics, Allegheny College May 2014

Publications

- Zoie T. Diana, Madeleine Milne, **Jacob Ford**, Ron Rubinovitz, Andrew Turner, Chelsea M. Rochman, "A Paint Library Of Plastic Particles (PLOPP): Fourier transform infrared spectral analysis of paint microplastics." *Microplastics and Nanoplastics*, September 2026.
- **Jacob Ford**, "Fairness in Focus: Insights into Bias Within Machine Learning Risk Evaluations and Established Credit Models." *Management System Engineering*, 2025.
- Zahra Thani, Winslow Lewis, Liz Neyens, **Jacob Ford**, "Priorities' Role in Community Solar: Survey-Based Study and Payment Performance Analysis." *Proceedings of the 52nd ASES National Solar Conference 2023*, Springer Proceedings in Energy, Springer, Cham, August 2023.
- Allison LaFave, Emily Kelly, **Jacob Ford**, "Reasons High School Students Change Their Educational Setting." *National Center for Education Statistics (NCES)*, NCES 2019-119, November 2018.
- Allison LaFave, Emily Kelly, **Jacob Ford**, "Factors that Influence Student College Choice." *National Center for Education Statistics (NCES)*, NCES 2018-119, June 2018.